Developing a Tool to Assist Digital 3D Model Balance and Animation

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Problem

Believable 3D animation depends heavily on the physical accuracy of motions. One of the major factors of believability is the **balance** of the poses of a 3D model. An unbalanced pose is easy to miss when animating but will subtly hurt the quality of the animation.

Proposed Solution

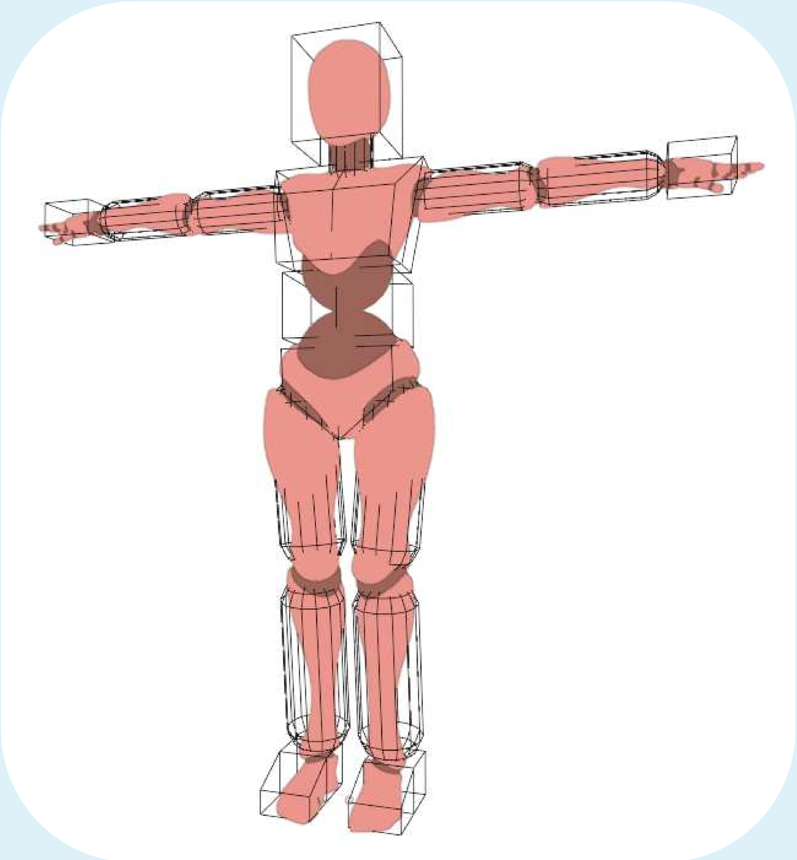
This problem was addressed with creation of a tool, native to Blender 3D, a free and powerful 3D tool suite. This will be done through **posing assistance**. Posing assistance will use properties about an animation rig and the body associated with it to determine if the pose the user created is balanced and assist in correction if it is not. This will greatly speed up the process of blocking it out animations, which is the first and most important part of the animation process.

Methodology

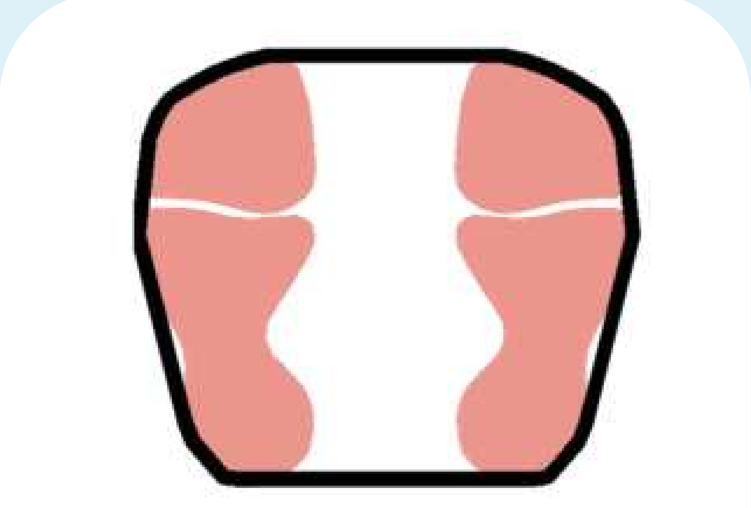
A "body of masses" was created representing the relative distribution of mass in the human body. This was used to calculate important physical information about the model to provide the user assistance with creating believable poses.

This body was used to find physical attributes for the model, such as the center of mass and the base of support (the area between the feet), which was used to determine if a pose was possible, allowing it to be corrected to improve animation quality.

This tool was accomplished through an add-on for Blender 3D. Blender natively supports python add-ons that can execute operations and take advantage of an API that allows access to many Blender features, allowing for the creation of the body of masses and each of the derivative features used for this tool.



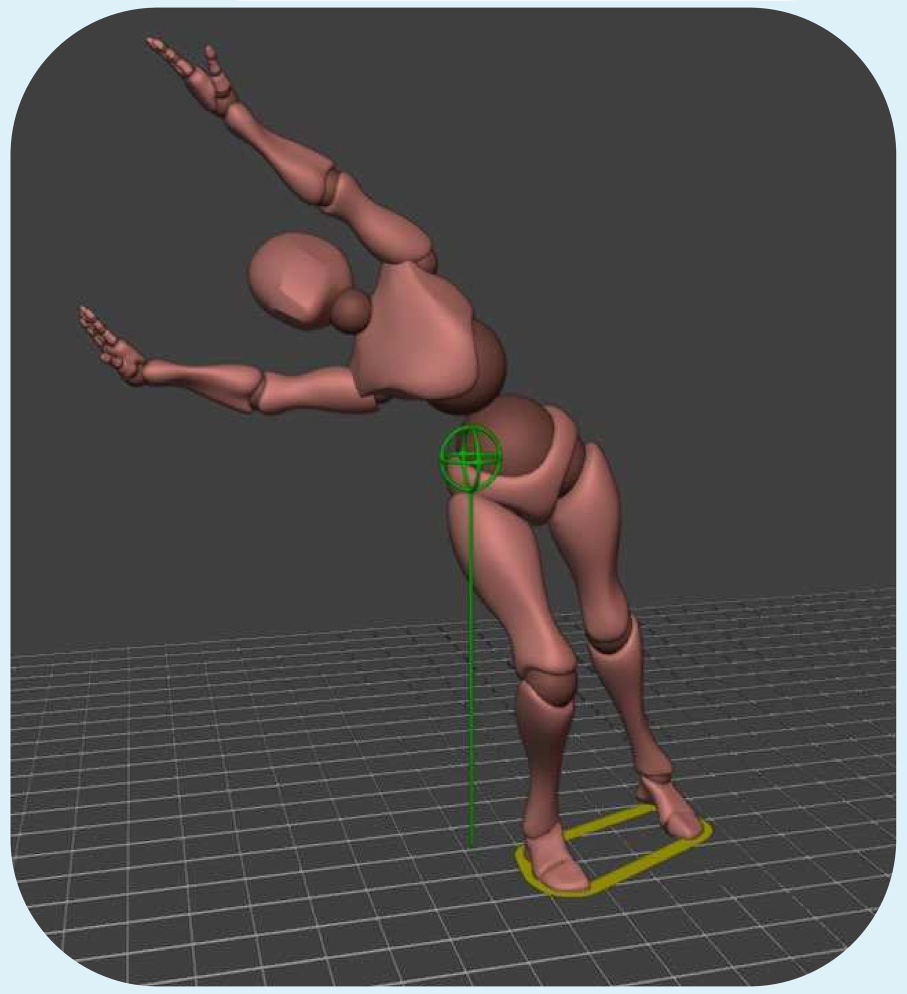
Body of Masses
Represented as wireframe over an example humanoid model.



Base of Support

Tool Capabilities

- Realistic Analogue for Human Body Mass Distribution
- Automatic Determination of Center of Mass
- Automatic Determination of Base of Support
- Automatic Determination of User Pose Balance
- Modular Design to Facilitate Customization and Expansion



An example of calculated information within Blender: he base of support, the center of mass, and its projection onto the ground.

Results

An add-on was developed for Blender 4.4.1 providing tools to assist the 3D animation with posing assistance tools. The add-on is still in progress, but it already features several useful tools, including those mentioned in Methodology as well as currently unused groundwork for expansion, such as tools and functions have been developed to easily and intuitively move bones within the rig.

Future Development

One of the priorities for this project was making many modular features to provide information on the physical attributes of a humanoid model. These tools form the groundwork for a system *automatically* correcting unbalanced poses. Beyond that, many tools for 3d animation would require the center of mass, base of support, physical information derived from the body of masses, so there is potential for expansion in other realms based on the information derived from this tool.